

1.5 Systems software

How to run this unit, lesson by lesson

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Start here

5 lessons. A realistic estimate is **5 teaching hours**: most lessons are one hour, and a lesson with thirty or more activities takes two or three. The estimate counts transitions and assumes you teach from the front for the first fifteen minutes of each lesson.

Teach them in the order below. The order is the retrieval schedule as well as the teaching one: each starter asks about the lesson before it.

#	LESSON	HOURS	PLAN
1	Control room ss-101	1	Lesson_Plans/SS_L01_LessonPlan.pdf
2	Multitasking floor ss-102	1	Lesson_Plans/SS_L02_LessonPlan.pdf
3	Accounts office ss-103	1	Lesson_Plans/SS_L03_LessonPlan.pdf
4	Utility workshop ss-104	1	Lesson_Plans/SS_L04_LessonPlan.pdf
5	End of topic test ss-105-test	1	Lesson_Plans/SS_L05_TEST_LessonPlan.pdf

Preparation and what you need

1. **Class logins, before the first lesson.** On the teacher dashboard (`teacher.html`), open **Class logins**, enter the class and the usernames one per line, and print the login cards. A pupil cannot sign in without one.
2. **Devices.** A laptop, a tablet or a phone will do. A headset is optional everywhere. The browser needs to be current; the experiences use WebGL.
3. **Print the worksheets** for each lesson. They are in this pack as .docx to edit and .pdf to print.
4. **Open each experience once yourself** on the teaching machine, so the images are cached before thirty pupils ask for them at the same moment.

Where progress is kept. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device a pupil used, not on a server. The teacher dashboard says so itself. Keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels with the pupil.

Pupils sign in at `topics.html` with the username and PIN on their card, choose the topic and then the lesson.

Using the three resources together

Slides for the front of the room; **the experience** for reading, deciding and being marked; **the worksheet** for writing and for the record. The rule that holds them together is write first, then answer at the badge.

Do not teach the station walls and then send pupils to read the same walls. Teach the first two or three from the slides and let them read the rest; the walls are there so a pupil can re-read what you said, not so you can say it twice.

Evidence, checking and common difficulties

Collect three things from every lesson: the starter (written), the key facts and station challenges on the worksheet (written), and the score screen breakdown (copied onto the worksheet by the pupil as soon as it appears). The score screen shows the marks per station, which is what tells you where to reteach; a single total does not.

The teacher dashboard shows progress by pupil and the class's weakest stations, averaged on first attempts. Use the weakest-station list to choose what to reteach rather than asking who found it hard.

Things to plan around in this unit

- every activity is guided: nothing is practised independently — ss-101
- no optional challenge activity — ss-101, ss-102, ss-103, ss-104

The lesson plans, in teaching order

Each lesson has its own plan as a separate PDF in Lesson_Plans/ of this pack, so you can print the one you are teaching. What follows is the summary of each; the plan carries the timings, the hinge question, the station-by-station delivery and the adaptations.

1. Control room ss-101

What does an operating system do, and why does every computer need one?

OUTCOMES	know the purpose and functionality of operating systems; know the different types of user interface and the features of each
STATIONS AND ACTIVITIES	6 stations, 13 activities, 28 marks (13 guided, 0 independent, 0 challenge)
RESOURCES	SS_L01_ControlRoom_Lesson.pptx, SS_L01_ControlRoom_Worksheet.docx
PLAN	Lesson_Plans/SS_L01_LessonPlan.pdf

2. Multitasking floor ss-102

How does an operating system share out the processor and the memory?

OUTCOMES	know what is meant by the term multitasking; understand how the operating system manages memory; understand the need for device drivers
STATIONS AND ACTIVITIES	6 stations, 14 activities, 22 marks (11 guided, 3 independent, 0 challenge)
RESOURCES	SS_L02_MultitaskingFloor_Lesson.pptx, SS_L02_MultitaskingFloor_Worksheet.docx
PLAN	Lesson_Plans/SS_L02_LessonPlan.pdf

3. Accounts office ss-103

How does an operating system manage users and their files?

OUTCOMES	understand what is meant by user management; understand the ways an operating system manages files
STATIONS AND ACTIVITIES	6 stations, 14 activities, 26 marks (12 guided, 2 independent, 0 challenge)
RESOURCES	SS_L03_AccountsOffice_Lesson.pptx, SS_L03_AccountsOffice_Worksheet.docx
PLAN	Lesson_Plans/SS_L03_LessonPlan.pdf

4. Utility workshop ss-104

What is the purpose of utility software?

OUTCOMES	understand encryption utilities; understand defragmentation utilities; understand data compression utilities
STATIONS AND ACTIVITIES	6 stations, 14 activities, 27 marks (12 guided, 2 independent, 0 challenge)

RESOURCES	SS_L04_UtilityWorkshop_Lesson.pptx, SS_L04_UtilityWorkshop_Worksheet.docx
PLAN	Lesson_Plans/SS_L04_LessonPlan.pdf

5. End of topic test ss-105-test

Your 1.5 test is done on paper. Revise with the four experiences first, then use this reflection sheet once it's marked.

OUTCOMES	this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.
STATIONS AND ACTIVITIES	0 stations, 0 activities, 0 marks (0 guided, 0 independent, 0 challenge)
RESOURCES	SS_L05_EndOfTopicTest_Reflection_Worksheet.docx
PLAN	Lesson_Plans/SS_L05_TEST_LessonPlan.pdf

Teacher answers

Model answers and marking guidance are not in this guide and not on the lesson plans. A plan is printed and left on a desk; an answer sheet is not. Where verified answers exist for this unit they are in the Teacher_Answers/ folder of this pack. Where they do not, the pack says so rather than shipping an empty file.

For the programming lessons there is no single model answer to print: a program is marked by running it against its test cases, and many different correct programs pass. The tests themselves are the marking guidance, and they are in the teacher answers.

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